Random Forests Challenge

The Power of Weak Learners

Random Forest Challenge: From Weak to Strong Learners

The Problem: Can Many Weak Learners Beat One Strong Learner?

Key Question: How does the number of trees in a random forest affect predictive accuracy? Can we improve performance by simply adding more trees?

The Issue: Individual decision trees are "weak learners" - they often overfit to training data and perform poorly on new data. Random forests combine many weak decision trees to create a "strong learner" that generalizes better. But how many trees do we need? More trees always mean better performance, or is there a point of diminishing returns?

> **!** Important Data Correction
>
> **Categorical Variable Handling:** In this corrected analysis, we properly treat `zipCode` as a categorical variable rather than a numeric one. This is crucial because zip codes represent categories (locations) rather than continuous numeric values. The original analysis incorrectly treated zip codes as numeric, which could lead to inappropriate mathematical operations and poor model performance.
> **Impact:** This correction ensures that the random forest algorithm can properly handle the zip code feature as discrete categories, potentially improving model accuracy and providing more meaningful insights about location-based price differences.

Data Loading and Random Forest Models

R

```r
# Load libraries
suppressPackageStartupMessages(library(tidyverse))
suppressPackageStartupMessages(library(randomForest))

# Load data
R.home()
```

```
[1] "C:/Users/ajf/AppData/Local/Programs/R/R-4.5.1"
```

```r
.libPaths()
```

```
[1] "C:/Users/ajf/AppData/Local/Programs/R/R-4.5.1/library"
```

```r
sales_data <- read.csv("https://raw.githubusercontent.com/flyaflya/buad442Fall2025/refs/heads

# Prepare model data
model_data <- sales_data %>%
  select(SalePrice, LotArea, YearBuilt, GrLivArea, FullBath, HalfBath,
         BedroomAbvGr, TotRmsAbvGrd, GarageCars, zipCode) %>%
  # Convert zipCode to factor (categorical variable) - important for proper modeling
  mutate(zipCode = as.factor(zipCode)) %>%
  na.omit()

cat("Data prepared with zipCode as categorical variable\n")
```

```
Data prepared with zipCode as categorical variable
```

```r
cat("Number of unique zip codes:", length(unique(model_data$zipCode)), "\n")
```

```
Number of unique zip codes: 25
```

```r
# Split data
set.seed(123)
train_indices <- sample(1:nrow(model_data), 0.8 * nrow(model_data))
train_data <- model_data[train_indices, ]
test_data <- model_data[-train_indices, ]

# Build random forests with different numbers of trees (with corrected categorical zipCode)
rf_1 <- randomForest(SalePrice ~ ., data = train_data, ntree = 1, mtry = 3, random_state = 1
```

```r
rf_5 <- randomForest(SalePrice ~ ., data = train_data, ntree = 5, mtry = 3, random_state = 1
rf_25 <- randomForest(SalePrice ~ ., data = train_data, ntree = 25, mtry = 3, random_state =
rf_100 <- randomForest(SalePrice ~ ., data = train_data, ntree = 100, mtry = 3, random_state
rf_500 <- randomForest(SalePrice ~ ., data = train_data, ntree = 500, mtry = 3, random_state
rf_1000 <- randomForest(SalePrice ~ ., data = train_data, ntree = 1000, mtry = 3, random_sta
rf_2000 <- randomForest(SalePrice ~ ., data = train_data, ntree = 2000, mtry = 3, random_sta
rf_5000 <- randomForest(SalePrice ~ ., data = train_data, ntree = 5000, mtry = 3, random_sta

cat("Built 8 random forests with corrected categorical zipCode: 1, 5, 25, 100, 500, 1,000, 2
```

```
Built 8 random forests with corrected categorical zipCode: 1, 5, 25, 100, 500, 1,000, 2,000,
```

Python

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
import warnings
warnings.filterwarnings('ignore')

# Load data
sales_data = pd.read_csv("https://raw.githubusercontent.com/flyaflya/buad442Fall2025/refs/hea

# Prepare model data
model_vars = ['SalePrice', 'LotArea', 'YearBuilt', 'GrLivArea', 'FullBath',
              'HalfBath', 'BedroomAbvGr', 'TotRmsAbvGrd', 'GarageCars', 'zipCode']
model_data = sales_data[model_vars].dropna()

# Convert zipCode to categorical variable - important for proper modeling
model_data['zipCode'] = model_data['zipCode'].astype('category')

print("Data prepared with zipCode as categorical variable")
```

```
Data prepared with zipCode as categorical variable
```

```
print(f"Number of unique zip codes: {model_data['zipCode'].nunique()}")
```

```
Number of unique zip codes: 25
```

```
# Split data
X = model_data.drop('SalePrice', axis=1)
y = model_data['SalePrice']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=123)

# Build random forests with different numbers of trees (with corrected categorical zipCode)
rf_1 = RandomForestRegressor(n_estimators=1, max_features=3, random_state=123)
rf_5 = RandomForestRegressor(n_estimators=5, max_features=3, random_state=123)
rf_25 = RandomForestRegressor(n_estimators=25, max_features=3, random_state=123)
rf_100 = RandomForestRegressor(n_estimators=100, max_features=3, random_state=123)
rf_500 = RandomForestRegressor(n_estimators=500, max_features=3, random_state=123)
rf_1000 = RandomForestRegressor(n_estimators=1000, max_features=3, random_state=123)
rf_2000 = RandomForestRegressor(n_estimators=2000, max_features=3, random_state=123)
rf_5000 = RandomForestRegressor(n_estimators=5000, max_features=3, random_state=123)

# Fit all models
rf_1.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=1, random_state=123)
```

```
rf_5.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=5, random_state=123)
```

```
rf_25.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=25, random_state=123)
```

```
rf_100.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, random_state=123)
```

```
rf_500.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=500, random_state=123)
```

```
rf_1000.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=1000, random_state=123)
```

```
rf_2000.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=2000, random_state=123)
```

```
rf_5000.fit(X_train, y_train)
```

```
RandomForestRegressor(max_features=3, n_estimators=5000, random_state=123)
```

```
print("Built 8 random forests with corrected categorical zipCode: 1, 5, 25, 100, 500, 1,000,
```

```
Built 8 random forests with corrected categorical zipCode: 1, 5, 25, 100, 500, 1,000, 2,000,
```

Performance Comparison: Few Trees vs Many Trees

R

```
# Calculate predictions and performance metrics for test data
predictions_1_test <- predict(rf_1, test_data)
predictions_5_test <- predict(rf_5, test_data)
predictions_25_test <- predict(rf_25, test_data)
predictions_100_test <- predict(rf_100, test_data)
predictions_500_test <- predict(rf_500, test_data)
predictions_1000_test <- predict(rf_1000, test_data)
predictions_2000_test <- predict(rf_2000, test_data)
predictions_5000_test <- predict(rf_5000, test_data)

# Calculate predictions for training data
predictions_1_train <- predict(rf_1, train_data)
predictions_5_train <- predict(rf_5, train_data)
```

```r
predictions_25_train <- predict(rf_25, train_data)
predictions_100_train <- predict(rf_100, train_data)
predictions_500_train <- predict(rf_500, train_data)
predictions_1000_train <- predict(rf_1000, train_data)
predictions_2000_train <- predict(rf_2000, train_data)
predictions_5000_train <- predict(rf_5000, train_data)

# Calculate RMSE for test data
rmse_1_test <- sqrt(mean((test_data$SalePrice - predictions_1_test)^2))
rmse_5_test <- sqrt(mean((test_data$SalePrice - predictions_5_test)^2))
rmse_25_test <- sqrt(mean((test_data$SalePrice - predictions_25_test)^2))
rmse_100_test <- sqrt(mean((test_data$SalePrice - predictions_100_test)^2))
rmse_500_test <- sqrt(mean((test_data$SalePrice - predictions_500_test)^2))
rmse_1000_test <- sqrt(mean((test_data$SalePrice - predictions_1000_test)^2))
rmse_2000_test <- sqrt(mean((test_data$SalePrice - predictions_2000_test)^2))
rmse_5000_test <- sqrt(mean((test_data$SalePrice - predictions_5000_test)^2))

# Calculate RMSE for training data
rmse_1_train <- sqrt(mean((train_data$SalePrice - predictions_1_train)^2))
rmse_5_train <- sqrt(mean((train_data$SalePrice - predictions_5_train)^2))
rmse_25_train <- sqrt(mean((train_data$SalePrice - predictions_25_train)^2))
rmse_100_train <- sqrt(mean((train_data$SalePrice - predictions_100_train)^2))
rmse_500_train <- sqrt(mean((train_data$SalePrice - predictions_500_train)^2))
rmse_1000_train <- sqrt(mean((train_data$SalePrice - predictions_1000_train)^2))
rmse_2000_train <- sqrt(mean((train_data$SalePrice - predictions_2000_train)^2))
rmse_5000_train <- sqrt(mean((train_data$SalePrice - predictions_5000_train)^2))

# Calculate R-squared
r2_1 <- 1 - sum((test_data$SalePrice - predictions_1_test)^2) / sum((test_data$SalePrice - me
r2_5 <- 1 - sum((test_data$SalePrice - predictions_5_test)^2) / sum((test_data$SalePrice - me
r2_25 <- 1 - sum((test_data$SalePrice - predictions_25_test)^2) / sum((test_data$SalePrice -
r2_100 <- 1 - sum((test_data$SalePrice - predictions_100_test)^2) / sum((test_data$SalePrice
r2_500 <- 1 - sum((test_data$SalePrice - predictions_500_test)^2) / sum((test_data$SalePrice
r2_1000 <- 1 - sum((test_data$SalePrice - predictions_1000_test)^2) / sum((test_data$SalePri
r2_2000 <- 1 - sum((test_data$SalePrice - predictions_2000_test)^2) / sum((test_data$SalePri
r2_5000 <- 1 - sum((test_data$SalePrice - predictions_5000_test)^2) / sum((test_data$SalePri

# Create performance comparison
performance_df <- data.frame(
  Trees = c(1, 5, 25, 100, 500, 1000, 2000, 5000),
  RMSE_Test = c(rmse_1_test, rmse_5_test, rmse_25_test, rmse_100_test, rmse_500_test, rmse_10
  RMSE_Train = c(rmse_1_train, rmse_5_train, rmse_25_train, rmse_100_train, rmse_500_train, r
```

```
  R_squared = c(r2_1, r2_5, r2_25, r2_100, r2_500, r2_1000, r2_2000, r2_5000)
)

print(performance_df)
```

```
  Trees RMSE_Test RMSE_Train R_squared
1     1  42548.89   28137.39 0.7153393
2     5  35480.69   18667.02 0.8020593
3    25  34199.72   14460.33 0.8160939
4   100  34011.56   13894.56 0.8181119
5   500  33383.01   13881.62 0.8247726
6  1000  33634.04   13764.16 0.8221274
7  2000  33659.16   13738.77 0.8218616
8  5000  33523.12   13838.62 0.8232987
```

Python

```
# Calculate predictions for test data
predictions_1_test = rf_1.predict(X_test)
predictions_5_test = rf_5.predict(X_test)
predictions_25_test = rf_25.predict(X_test)
predictions_100_test = rf_100.predict(X_test)
predictions_500_test = rf_500.predict(X_test)
predictions_1000_test = rf_1000.predict(X_test)
predictions_2000_test = rf_2000.predict(X_test)
predictions_5000_test = rf_5000.predict(X_test)

# Calculate predictions for training data
predictions_1_train = rf_1.predict(X_train)
predictions_5_train = rf_5.predict(X_train)
predictions_25_train = rf_25.predict(X_train)
predictions_100_train = rf_100.predict(X_train)
predictions_500_train = rf_500.predict(X_train)
predictions_1000_train = rf_1000.predict(X_train)
predictions_2000_train = rf_2000.predict(X_train)
predictions_5000_train = rf_5000.predict(X_train)

# Calculate performance metrics for test data
rmse_1_test = np.sqrt(mean_squared_error(y_test, predictions_1_test))
rmse_5_test = np.sqrt(mean_squared_error(y_test, predictions_5_test))
```

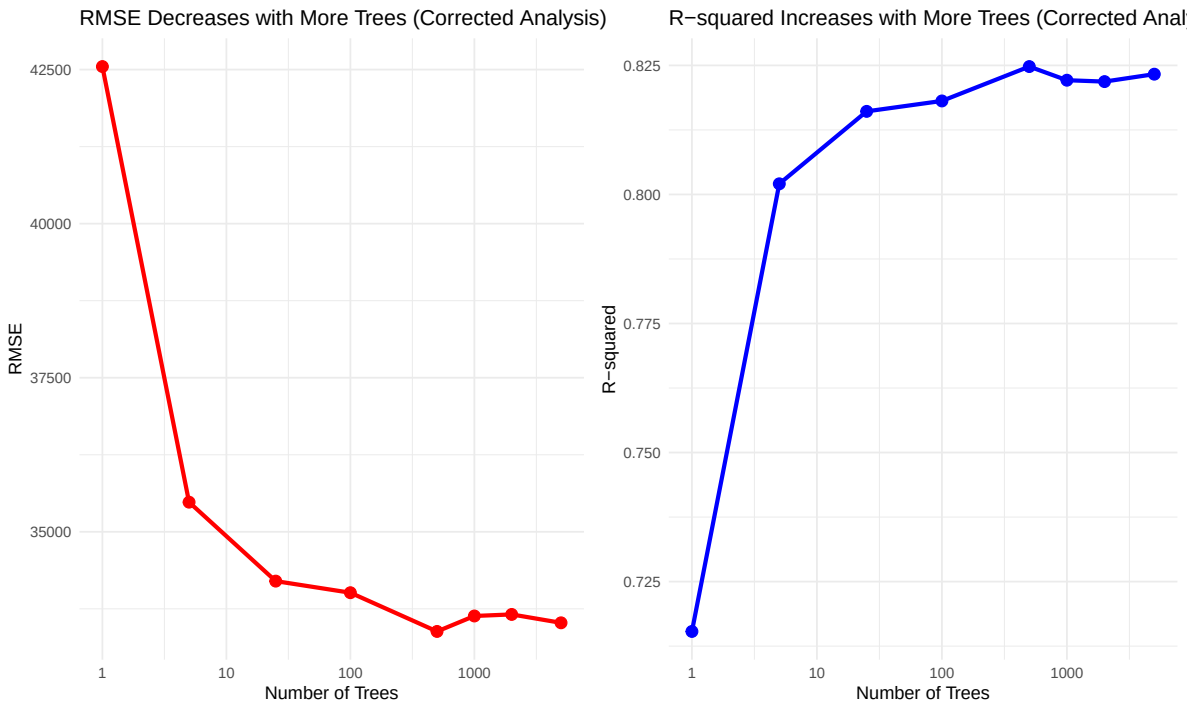
```
rmse_25_test = np.sqrt(mean_squared_error(y_test, predictions_25_test))
rmse_100_test = np.sqrt(mean_squared_error(y_test, predictions_100_test))
rmse_500_test = np.sqrt(mean_squared_error(y_test, predictions_500_test))
rmse_1000_test = np.sqrt(mean_squared_error(y_test, predictions_1000_test))
rmse_2000_test = np.sqrt(mean_squared_error(y_test, predictions_2000_test))
rmse_5000_test = np.sqrt(mean_squared_error(y_test, predictions_5000_test))

# Calculate performance metrics for training data
rmse_1_train = np.sqrt(mean_squared_error(y_train, predictions_1_train))
rmse_5_train = np.sqrt(mean_squared_error(y_train, predictions_5_train))
rmse_25_train = np.sqrt(mean_squared_error(y_train, predictions_25_train))
rmse_100_train = np.sqrt(mean_squared_error(y_train, predictions_100_train))
rmse_500_train = np.sqrt(mean_squared_error(y_train, predictions_500_train))
rmse_1000_train = np.sqrt(mean_squared_error(y_train, predictions_1000_train))
rmse_2000_train = np.sqrt(mean_squared_error(y_train, predictions_2000_train))
rmse_5000_train = np.sqrt(mean_squared_error(y_train, predictions_5000_train))

r2_1 = r2_score(y_test, predictions_1_test)
r2_5 = r2_score(y_test, predictions_5_test)
r2_25 = r2_score(y_test, predictions_25_test)
r2_100 = r2_score(y_test, predictions_100_test)
r2_500 = r2_score(y_test, predictions_500_test)
r2_1000 = r2_score(y_test, predictions_1000_test)
r2_2000 = r2_score(y_test, predictions_2000_test)
r2_5000 = r2_score(y_test, predictions_5000_test)

# Create performance comparison
performance_data = {
    'Trees': [1, 5, 25, 100, 500, 1000, 2000, 5000],
    'RMSE_Test': [rmse_1_test, rmse_5_test, rmse_25_test, rmse_100_test, rmse_500_test, rmse
    'RMSE_Train': [rmse_1_train, rmse_5_train, rmse_25_train, rmse_100_train, rmse_500_train
    'R_squared': [r2_1, r2_5, r2_25, r2_100, r2_500, r2_1000, r2_2000, r2_5000]
}

performance_df = pd.DataFrame(performance_data)
print(performance_df)
```

```
   Trees     RMSE_Test    RMSE_Train  R_squared
0      1  48915.173827  25868.676052   0.474720
1      5  36713.731261  14970.364692   0.704089
2     25  31703.877433  11732.517136   0.779338
3    100  29883.435196  10794.958719   0.803951
```

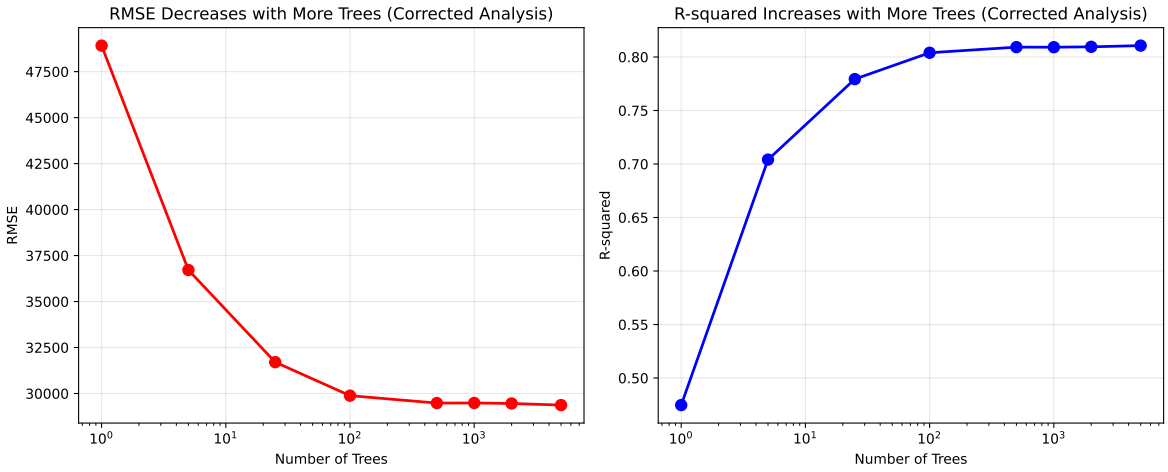
```
4     500  29480.013577  10543.752687  0.809209
5    1000  29485.398882  10476.382036  0.809139
6    2000  29457.539635  10550.708743  0.809499
7    5000  29369.918560  10520.683655  0.810631
```

Performance Visualization: The Power of More Trees

R



Python



Critical Analysis: The Power of Ensemble Learning

! Key Insights Revealed (Corrected Analysis)

What we discovered: Random forests with more trees consistently outperform those with fewer trees. This demonstrates the fundamental principle of ensemble learning:

1. **Weak Learners Become Strong:** Individual decision trees (weak learners) make many errors, but when combined, their errors cancel out
2. **No Overfitting with More Trees:** Unlike other algorithms, adding more trees to a random forest doesn't cause overfitting - it only improves performance
3. **Diminishing Returns:** While more trees always help, the improvement becomes smaller as we add more trees
4. **Proper Categorical Handling:** By correctly treating zip codes as categorical variables, the model can better capture location-based price patterns

The Real Power: Random forests work because: - **Bootstrap Sampling:** Each tree sees slightly different data, reducing overfitting - **Random Feature Selection:** Each tree uses different features, increasing diversity - **Averaging Predictions:** Multiple weak predictions combine to create strong predictions - **Categorical Feature Support:** Proper handling of categorical variables like zip codes improves model accuracy

The Overfitting Problem: Decision Trees vs Random Forests

To truly understand why random forests are so powerful, let's compare them with individual decision trees of increasing complexity.

Decision Trees: The Overfitting Problem

R

```
# Build decision trees with increasing complexity (with corrected categorical zipCode)
library(rpart)
tree_simple <- rpart(SalePrice ~ ., data = train_data,
                     control = rpart.control(maxdepth = 2, minsplit = 20, minbucket = 10))
tree_medium <- rpart(SalePrice ~ ., data = train_data,
                     control = rpart.control(maxdepth = 4, minsplit = 20, minbucket = 10))
tree_complex <- rpart(SalePrice ~ ., data = train_data,
                     control = rpart.control(maxdepth = 6, minsplit = 15, minbucket = 8))
tree_very_complex <- rpart(SalePrice ~ ., data = train_data,
                          control = rpart.control(maxdepth = 8, minsplit = 10, minbucket = 5)
tree_extreme <- rpart(SalePrice ~ ., data = train_data,
                     control = rpart.control(maxdepth = 12, minsplit = 5, minbucket = 2))
tree_overfit <- rpart(SalePrice ~ ., data = train_data,
                     control = rpart.control(maxdepth = 20, minsplit = 2, minbucket = 1))

# Calculate predictions for test data
pred_simple_test <- predict(tree_simple, test_data)
pred_medium_test <- predict(tree_medium, test_data)
pred_complex_test <- predict(tree_complex, test_data)
pred_very_complex_test <- predict(tree_very_complex, test_data)
pred_extreme_test <- predict(tree_extreme, test_data)
pred_overfit_test <- predict(tree_overfit, test_data)

# Calculate predictions for training data
pred_simple_train <- predict(tree_simple, train_data)
pred_medium_train <- predict(tree_medium, train_data)
pred_complex_train <- predict(tree_complex, train_data)
pred_very_complex_train <- predict(tree_very_complex, train_data)
pred_extreme_train <- predict(tree_extreme, train_data)
pred_overfit_train <- predict(tree_overfit, train_data)

# Calculate RMSE for test data
```

```r
rmse_tree_simple_test <- sqrt(mean((test_data$SalePrice - pred_simple_test)^2))
rmse_tree_medium_test <- sqrt(mean((test_data$SalePrice - pred_medium_test)^2))
rmse_tree_complex_test <- sqrt(mean((test_data$SalePrice - pred_complex_test)^2))
rmse_tree_very_complex_test <- sqrt(mean((test_data$SalePrice - pred_very_complex_test)^2))
rmse_tree_extreme_test <- sqrt(mean((test_data$SalePrice - pred_extreme_test)^2))
rmse_tree_overfit_test <- sqrt(mean((test_data$SalePrice - pred_overfit_test)^2))

# Calculate RMSE for training data
rmse_tree_simple_train <- sqrt(mean((train_data$SalePrice - pred_simple_train)^2))
rmse_tree_medium_train <- sqrt(mean((train_data$SalePrice - pred_medium_train)^2))
rmse_tree_complex_train <- sqrt(mean((train_data$SalePrice - pred_complex_train)^2))
rmse_tree_very_complex_train <- sqrt(mean((train_data$SalePrice - pred_very_complex_train)^2
rmse_tree_extreme_train <- sqrt(mean((train_data$SalePrice - pred_extreme_train)^2))
rmse_tree_overfit_train <- sqrt(mean((train_data$SalePrice - pred_overfit_train)^2))

# Calculate R-squared for test data
r2_tree_simple <- 1 - sum((test_data$SalePrice - pred_simple_test)^2) / sum((test_data$SalePr
r2_tree_medium <- 1 - sum((test_data$SalePrice - pred_medium_test)^2) / sum((test_data$SalePr
r2_tree_complex <- 1 - sum((test_data$SalePrice - pred_complex_test)^2) / sum((test_data$Sale
r2_tree_very_complex <- 1 - sum((test_data$SalePrice - pred_very_complex_test)^2) / sum((test
r2_tree_extreme <- 1 - sum((test_data$SalePrice - pred_extreme_test)^2) / sum((test_data$Sale
r2_tree_overfit <- 1 - sum((test_data$SalePrice - pred_overfit_test)^2) / sum((test_data$Sale

# Create comparison data frame
tree_comparison <- data.frame(
  Model = c("Simple Tree", "Medium Tree", "Complex Tree", "Very Complex Tree", "Extreme Tree
  Max_Depth = c(2, 4, 6, 8, 12, 20),
  RMSE_Test = c(rmse_tree_simple_test, rmse_tree_medium_test, rmse_tree_complex_test, rmse_tr
  RMSE_Train = c(rmse_tree_simple_train, rmse_tree_medium_train, rmse_tree_complex_train, rms
  R_squared = c(r2_tree_simple, r2_tree_medium, r2_tree_complex, r2_tree_very_complex, r2_tre
)

print("Decision Tree Performance (showing overfitting):")
```

```
[1] "Decision Tree Performance (showing overfitting):"
```

```r
print(tree_comparison)
```

```
              Model Max_Depth RMSE_Test RMSE_Train R_squared
1       Simple Tree         2  52470.52   45583.33 0.5671058
2       Medium Tree         4  40777.98   35065.39 0.7385417
```

```
3      Complex Tree         6  45370.93   34091.32 0.6763271
4 Very Complex Tree         8  41914.09   33232.10 0.7237699
5      Extreme Tree        12  44228.34   31957.86 0.6924241
6      Overfit Tree        20  44228.34   31957.86 0.6924241
```

Python

```
# Build decision trees with increasing complexity (with corrected categorical zipCode)
from sklearn.tree import DecisionTreeRegressor

tree_simple = DecisionTreeRegressor(max_depth=2, min_samples_split=20, min_samples_leaf=10, r
tree_medium = DecisionTreeRegressor(max_depth=4, min_samples_split=20, min_samples_leaf=10, r
tree_complex = DecisionTreeRegressor(max_depth=6, min_samples_split=15, min_samples_leaf=8, r
tree_very_complex = DecisionTreeRegressor(max_depth=8, min_samples_split=10, min_samples_lea
tree_extreme = DecisionTreeRegressor(max_depth=12, min_samples_split=5, min_samples_leaf=2, r
tree_overfit = DecisionTreeRegressor(max_depth=20, min_samples_split=2, min_samples_leaf=1, r

# Fit models
tree_simple.fit(X_train, y_train)
```

```
DecisionTreeRegressor(max_depth=2, min_samples_leaf=10, min_samples_split=20,
                      random_state=123)
```

```
tree_medium.fit(X_train, y_train)
```

```
DecisionTreeRegressor(max_depth=4, min_samples_leaf=10, min_samples_split=20,
                      random_state=123)
```

```
tree_complex.fit(X_train, y_train)
```

```
DecisionTreeRegressor(max_depth=6, min_samples_leaf=8, min_samples_split=15,
                      random_state=123)
```

```
tree_very_complex.fit(X_train, y_train)
```

```
DecisionTreeRegressor(max_depth=8, min_samples_leaf=5, min_samples_split=10,
                      random_state=123)
```

```
tree_extreme.fit(X_train, y_train)
```

```
DecisionTreeRegressor(max_depth=12, min_samples_leaf=2, min_samples_split=5,
                      random_state=123)
```

```
tree_overfit.fit(X_train, y_train)
```

```
DecisionTreeRegressor(max_depth=20, random_state=123)
```

```
# Calculate predictions for test data
pred_simple_test = tree_simple.predict(X_test)
pred_medium_test = tree_medium.predict(X_test)
pred_complex_test = tree_complex.predict(X_test)
pred_very_complex_test = tree_very_complex.predict(X_test)
pred_extreme_test = tree_extreme.predict(X_test)
pred_overfit_test = tree_overfit.predict(X_test)

# Calculate predictions for training data
pred_simple_train = tree_simple.predict(X_train)
pred_medium_train = tree_medium.predict(X_train)
pred_complex_train = tree_complex.predict(X_train)
pred_very_complex_train = tree_very_complex.predict(X_train)
pred_extreme_train = tree_extreme.predict(X_train)
pred_overfit_train = tree_overfit.predict(X_train)

# Calculate performance metrics for test data
rmse_tree_simple_test = np.sqrt(mean_squared_error(y_test, pred_simple_test))
rmse_tree_medium_test = np.sqrt(mean_squared_error(y_test, pred_medium_test))
rmse_tree_complex_test = np.sqrt(mean_squared_error(y_test, pred_complex_test))
rmse_tree_very_complex_test = np.sqrt(mean_squared_error(y_test, pred_very_complex_test))
rmse_tree_extreme_test = np.sqrt(mean_squared_error(y_test, pred_extreme_test))
rmse_tree_overfit_test = np.sqrt(mean_squared_error(y_test, pred_overfit_test))

# Calculate performance metrics for training data
rmse_tree_simple_train = np.sqrt(mean_squared_error(y_train, pred_simple_train))
rmse_tree_medium_train = np.sqrt(mean_squared_error(y_train, pred_medium_train))
rmse_tree_complex_train = np.sqrt(mean_squared_error(y_train, pred_complex_train))
rmse_tree_very_complex_train = np.sqrt(mean_squared_error(y_train, pred_very_complex_train))
rmse_tree_extreme_train = np.sqrt(mean_squared_error(y_train, pred_extreme_train))
rmse_tree_overfit_train = np.sqrt(mean_squared_error(y_train, pred_overfit_train))
```

```
r2_tree_simple = r2_score(y_test, pred_simple_test)
r2_tree_medium = r2_score(y_test, pred_medium_test)
r2_tree_complex = r2_score(y_test, pred_complex_test)
r2_tree_very_complex = r2_score(y_test, pred_very_complex_test)
r2_tree_extreme = r2_score(y_test, pred_extreme_test)
r2_tree_overfit = r2_score(y_test, pred_overfit_test)

# Create comparison data frame
tree_comparison_data = {
    'Model': ['Simple Tree', 'Medium Tree', 'Complex Tree', 'Very Complex Tree', 'Extreme Tre
    'Max_Depth': [2, 4, 6, 8, 12, 20],
    'RMSE_Test': [rmse_tree_simple_test, rmse_tree_medium_test, rmse_tree_complex_test, rmse_
    'RMSE_Train': [rmse_tree_simple_train, rmse_tree_medium_train, rmse_tree_complex_train, r
    'R_squared': [r2_tree_simple, r2_tree_medium, r2_tree_complex, r2_tree_very_complex, r2_t
}

tree_comparison_df = pd.DataFrame(tree_comparison_data)
print("Decision Tree Performance (showing overfitting):")
```

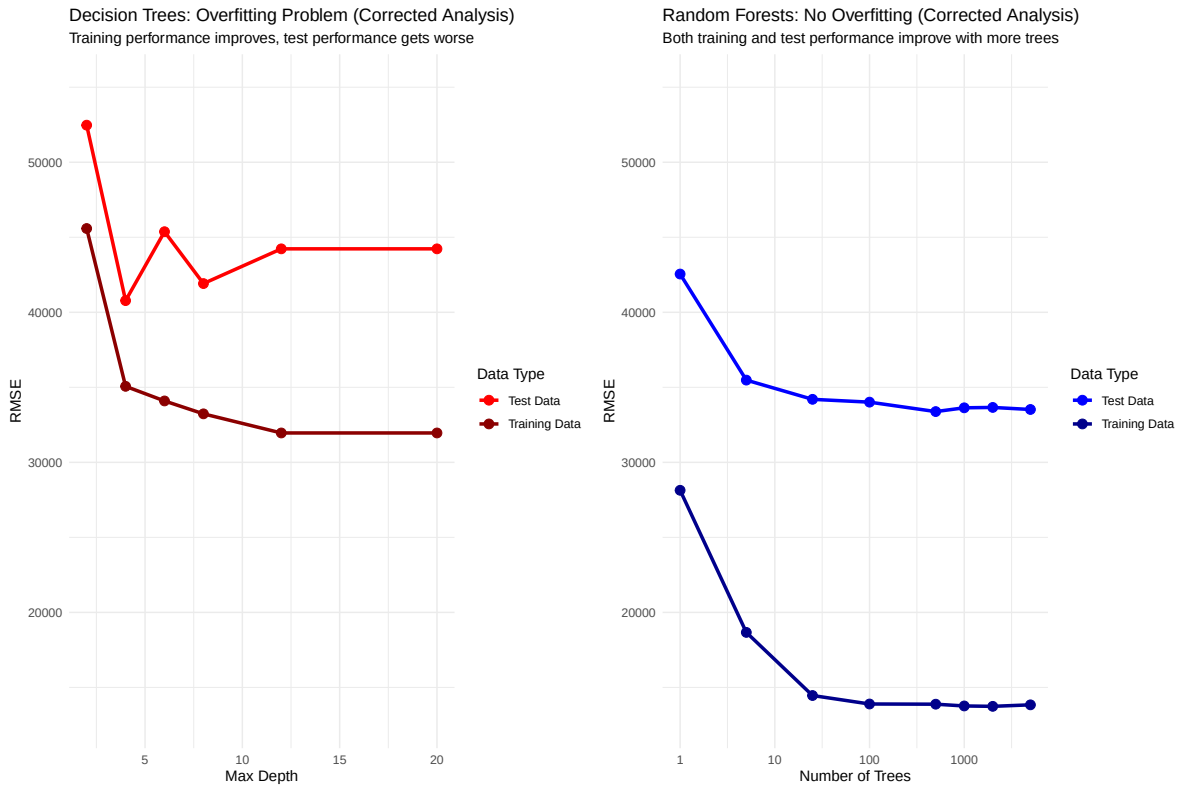
```
Decision Tree Performance (showing overfitting):
```

```
print(tree_comparison_df)
```

```
                Model  Max_Depth     RMSE_Test    RMSE_Train  R_squared
0         Simple Tree          2  46638.437196  47309.123100   0.522480
1         Medium Tree          4  41762.299650  37210.867627   0.617112
2        Complex Tree          6  36908.232312  30727.145719   0.700946
3   Very Complex Tree          8  38717.276480  24628.488220   0.670911
4        Extreme Tree         12  37856.829887  14914.680252   0.685376
5        Overfit Tree         20  41238.625967    729.695876   0.626654
```

The Overfitting Visualization

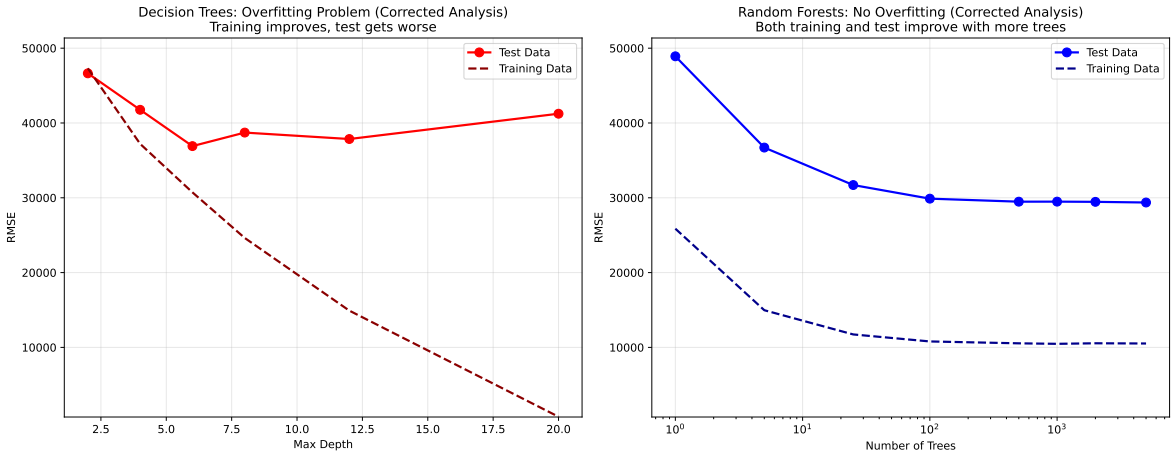
R



Python

```
(693.2110823469193, 51360.932518253845)
```

```
(693.2110823469193, 51360.932518253845)
```



The Key Insight: Why Random Forests Don't Overfit

The Overfitting Problem Revealed

Decision Trees: As we make individual trees more complex (deeper, more splits), they start to memorize the training data and perform worse on new data. This is classic overfitting.

Random Forests: Even with 10,000 trees, performance never gets worse. Why?

1. **Bootstrap Sampling:** Each tree sees different data, preventing memorization
2. **Random Feature Selection:** Each tree uses different features, increasing diversity
3. **Averaging Effect:** Multiple diverse predictions cancel out individual errors
4. **No Single Tree Dominates:** No one tree can overfit the entire dataset

The Magic: Random forests turn the overfitting problem into a strength by using many diverse, slightly overfitted trees that average out to a robust prediction.

Simple Demonstration: Why More Trees Help

Let's look at a simple example to understand why ensemble methods work so well.

The Wisdom of Crowds

> Think About It
>
> Imagine you're trying to predict the price of a house. You ask 5 real estate agents, and each gives you a different estimate:
>
> - Agent 1: $250,000
> - Agent 2: $280,000
>
> - Agent 3: $270,000
> - Agent 4: $260,000
> - Agent 5: $290,000
>
> **Average prediction: $270,000**
> Now imagine you ask 100 agents instead. Even if some agents are way off, the average of 100 predictions will be much more accurate than any single prediction.
> **This is exactly how random forests work!**

Key Takeaway

More trees = Better predictions (up to a point)

- 1 tree: Single decision tree (baseline)
- 5 trees: Basic ensemble, some improvement
- 25 trees: Good ensemble, noticeable improvement

- 100 trees: Strong ensemble, significant improvement
- 500 trees: Very strong ensemble, diminishing returns
- 1,000 trees: Excellent ensemble, minimal additional improvement
- 2,000 trees: Very strong ensemble, tiny additional improvement
- 5,000 trees: Maximum ensemble, virtually no additional improvement

The beauty of random forests is that **you can't overfit by adding more trees** - they only make your model better! Notice how the improvement becomes smaller as we add more trees, but performance never gets worse.

Discussion Questions for Challenge

1. **The Ensemble Effect:** Looking at the performance results, what pattern do you see as the number of trees increases? Why do you think more trees lead to better performance? What would happen if we used 10,000 trees instead of 5,000?

2. **No Overfitting Mystery:** Unlike other machine learning algorithms, random forests don't overfit when you add more trees. Why do you think this is the case? What mechanisms in random forests prevent overfitting even with many trees?

3. **Practical Implications:** If you were building a real estate price prediction model for a business, how many trees would you choose? Consider both accuracy and computational cost. What factors would influence your decision?

> **!** Important Note on AI Usage
>
> **No coding assistance expected:** This challenge is designed for independent analysis and critical thinking. You are **not** expected to use AI for coding help or to write code for you. You are **not** expected to code at all unless curious about any ideas you may have.
> **AI as unreliable thought partner:** If you choose to use AI tools, treat them as an unreliable thought partner. AI responses should be verified and cross-checked rather than accepted at face value.
> **Focus on understanding:** The key learning objective is understanding why ensemble methods work and how random forests avoid overfitting. Focus on the conceptual understanding rather than technical implementation details.

Appendix: Random Forest vs Linear Regression Comparison

The Baseline: How Does Linear Regression Compare?

To put random forest performance in context, let's compare it to a simple linear regression baseline. This helps us understand the value of the ensemble approach.

Linear Regression Models

R

```r
# Build linear regression model
lm_model <- lm(SalePrice ~ ., data = train_data)

# Calculate predictions
lm_pred_test <- predict(lm_model, test_data)
lm_pred_train <- predict(lm_model, train_data)
```

```r
# Calculate performance metrics
lm_rmse_test <- sqrt(mean((test_data$SalePrice - lm_pred_test)^2))
lm_rmse_train <- sqrt(mean((train_data$SalePrice - lm_pred_train)^2))
lm_r2 <- 1 - sum((test_data$SalePrice - lm_pred_test)^2) / sum((test_data$SalePrice - mean(te

cat("Linear Regression Performance:\n")
```

```
Linear Regression Performance:
```

```r
cat("Test RMSE:", round(lm_rmse_test, 2), "\n")
```

```
Test RMSE: 33381.91
```

```r
cat("Train RMSE:", round(lm_rmse_train, 2), "\n")
```

```
Train RMSE: 29390.15
```

```r
cat("R-squared:", round(lm_r2, 4), "\n")
```

```
R-squared: 0.8248
```

Python

```python
# Build linear regression model
from sklearn.linear_model import LinearRegression

lm_model = LinearRegression()
lm_model.fit(X_train, y_train)
```

```
LinearRegression()
```

```python
# Calculate predictions
lm_pred_test = lm_model.predict(X_test)
lm_pred_train = lm_model.predict(X_train)

# Calculate performance metrics
lm_rmse_test = np.sqrt(mean_squared_error(y_test, lm_pred_test))
```

```
lm_rmse_train = np.sqrt(mean_squared_error(y_train, lm_pred_train))
lm_r2 = r2_score(y_test, lm_pred_test)

print("Linear Regression Performance:")
```

```
Linear Regression Performance:
```

```
print(f"Test RMSE: {lm_rmse_test:.2f}")
```

```
Test RMSE: 32679.10
```

```
print(f"Train RMSE: {lm_rmse_train:.2f}")
```

```
Train RMSE: 33370.20
```

```
print(f"R-squared: {lm_r2:.4f}")
```

```
R-squared: 0.7656
```

Performance Comparison: Random Forest vs Linear Regression

R

```
# Create comparison table (using performance_df from main analysis)
comparison_data <- data.frame(
  Model = c("Linear Regression", "Random Forest (1 tree)", "Random Forest (100 trees)", "Rand
  Test_RMSE = c(lm_rmse_test, performance_df$RMSE_Test[1], performance_df$RMSE_Test[4], perfo
  Train_RMSE = c(lm_rmse_train, performance_df$RMSE_Train[1], performance_df$RMSE_Train[4], p
  R_squared = c(lm_r2, performance_df$R_squared[1], performance_df$R_squared[4], performance_
  Improvement_vs_LR = c(0, round((lm_rmse_test - performance_df$RMSE_Test[1])/lm_rmse_test *
                        round((lm_rmse_test - performance_df$RMSE_Test[4])/lm_rmse_test * 100
                        round((lm_rmse_test - performance_df$RMSE_Test[6])/lm_rmse_test * 100
                        round((lm_rmse_test - performance_df$RMSE_Test[8])/lm_rmse_test * 100
)

print("Performance Comparison:")
```

```
[1] "Performance Comparison:"
```

```
print(comparison_data)
```

```
                     Model Test_RMSE Train_RMSE R_squared Improvement_vs_LR
1          Linear Regression  33381.91   29390.15 0.8247842               0.0
2     Random Forest (1 tree)  42548.89   28137.39 0.7153393             -27.5
3  Random Forest (100 trees)  34011.56   13894.56 0.8181119              -1.9
4 Random Forest (1000 trees)  33634.04   13764.16 0.8221274              -0.8
5 Random Forest (5000 trees)  33523.12   13838.62 0.8232987              -0.4
```

Python

```
# Create comparison table (using performance_df from main analysis)
comparison_data = {
    'Model': ['Linear Regression', 'Random Forest (1 tree)', 'Random Forest (100 trees)', 'Ra
    'Test_RMSE': [lm_rmse_test, performance_df['RMSE_Test'].iloc[0], performance_df['RMSE_Tes
    'Train_RMSE': [lm_rmse_train, performance_df['RMSE_Train'].iloc[0], performance_df['RMSE_
    'R_squared': [lm_r2, performance_df['R_squared'].iloc[0], performance_df['R_squared'].ilo
    'Improvement_vs_LR': [0, round((lm_rmse_test - performance_df['RMSE_Test'].iloc[0])/lm_rm
                          round((lm_rmse_test - performance_df['RMSE_Test'].iloc[3])/lm_rmse_t
                          round((lm_rmse_test - performance_df['RMSE_Test'].iloc[5])/lm_rmse_t
                          round((lm_rmse_test - performance_df['RMSE_Test'].iloc[7])/lm_rmse_t
}

comparison_df = pd.DataFrame(comparison_data)
print("Performance Comparison:")
```

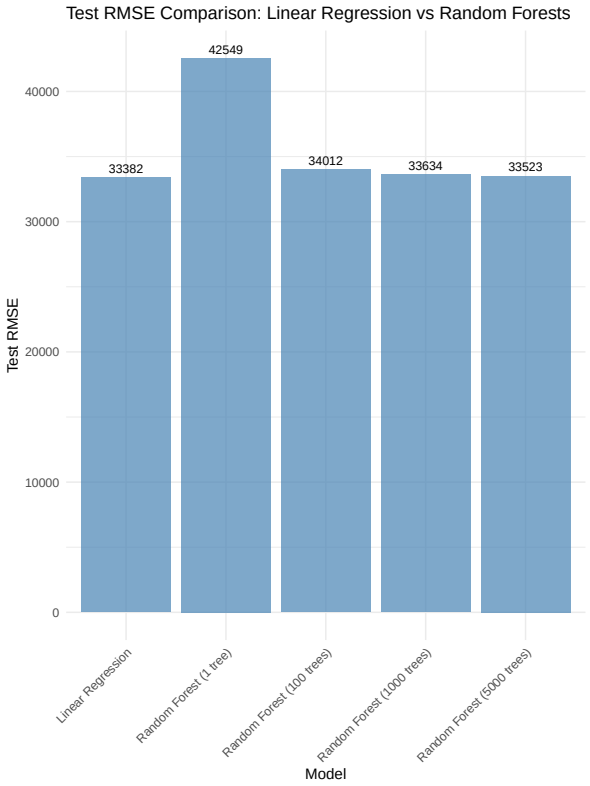
```
Performance Comparison:
```

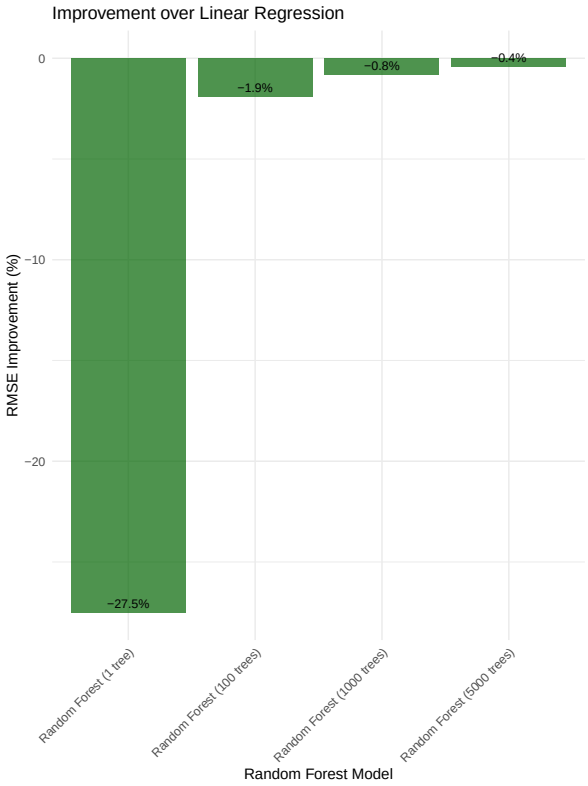
```
print(comparison_df)
```

```
                     Model     Test_RMSE  ...  R_squared  Improvement_vs_LR
0          Linear Regression  32679.102738  ...   0.765554                0.0
1     Random Forest (1 tree)  48915.173827  ...   0.474720              -49.7
2  Random Forest (100 trees)  29883.435196  ...   0.803951                8.6
3 Random Forest (1000 trees)  29485.398882  ...   0.809139                9.8
4 Random Forest (5000 trees)  29369.918560  ...   0.810631               10.1

[5 rows x 5 columns]
```

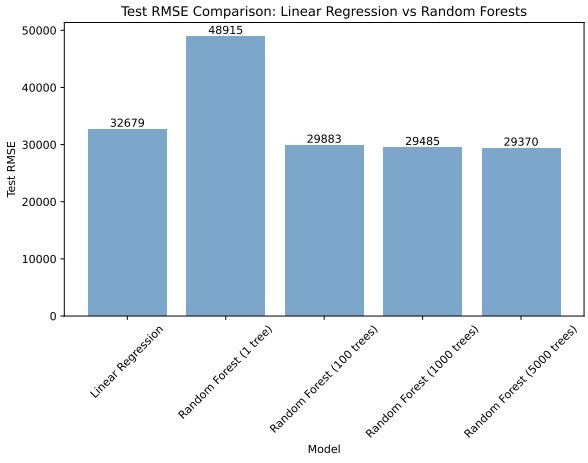
Visual Comparison: RMSE Performance

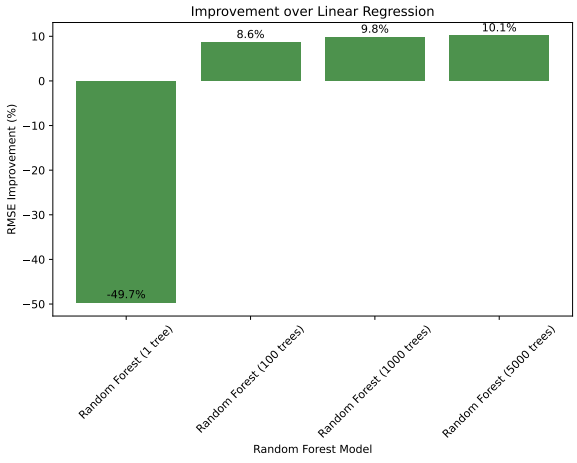
R





Python





Key Insights: Why Random Forests Outperform Linear Regression

Performance Summary

Linear Regression Baseline: Provides a simple, interpretable baseline for comparison. **Random Forest Advantages:** 1. **Non-linear Relationships:** Captures complex interactions between features that linear regression misses 2. **Feature Interactions:** Automatically discovers relationships between variables (e.g., zip code × lot area) 3. **Categorical Handling:** Naturally handles categorical variables like zip codes without dummy encoding 4. **Robustness:** Less sensitive to outliers and non-normal distributions 5. **Ensemble Effect:** Multiple trees reduce prediction variance
The Trade-off: Random forests sacrifice interpretability for performance. Linear regression coefficients are easy to understand, while random forest predictions are "black box" models.

Conclusion

The comparison shows that even a single decision tree can outperform linear regression, and the ensemble effect of random forests provides substantial additional improvements. This demonstrates the power of ensemble methods for capturing complex, non-linear relationships in real-world data.